

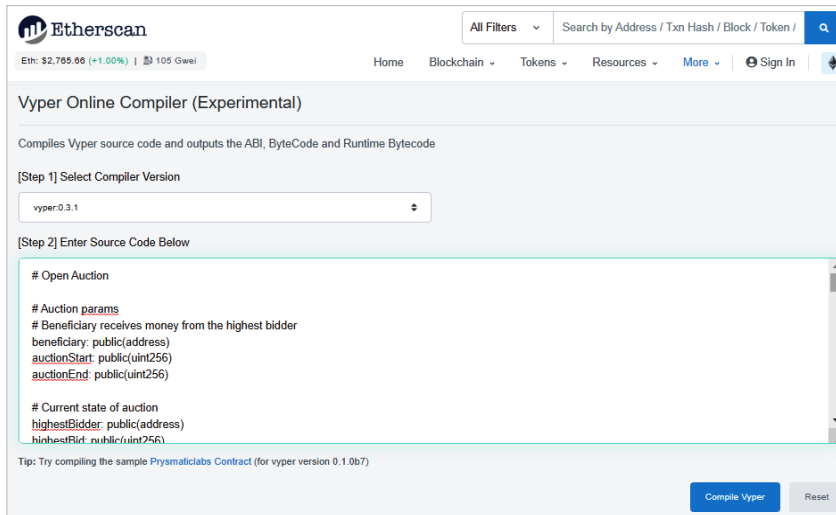


Figure 4: Online compiler for Vyper based smart contracts

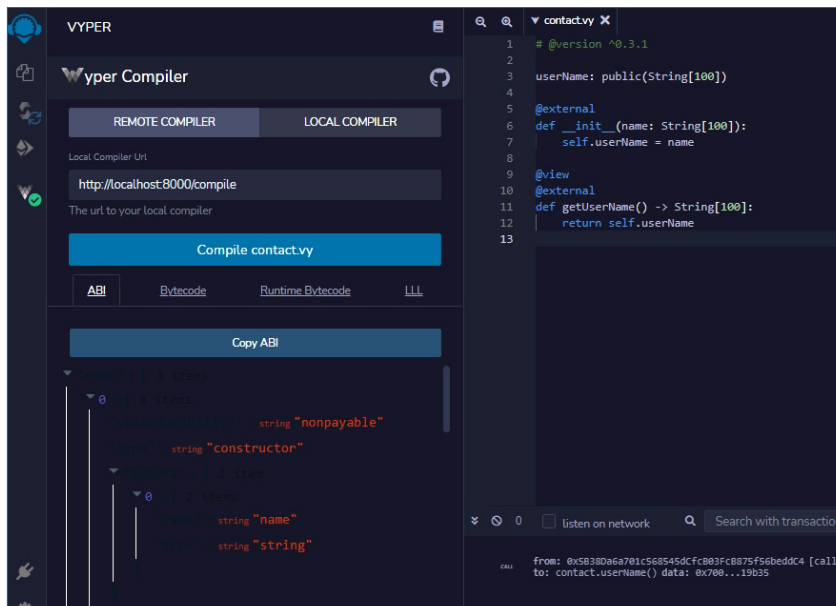


Figure 5: Compiling and running a smart contract using Vyper

variety of settings and platforms for use with blockchains. Numerous IDEs and editors, such as Remix, EthFiddle, JetBrains, and many more, are available for use with the Solidity programming language.

Remix IDE can be used for the development and deployment of smart contracts so that high performance decentralized apps can be deployed on blockchain (Figure 1).

This IDE integrates an effective and easy to use graphical user interface

(GUI) for blockchain developers so that smart contracts can be written, compiled and deployed easily (Figure 2).

After writing and compiling a smart contract in Remix IDE, the Ethereum blockchain can be deployed; its log records are available on the Ethereum platform (Figure 3).

The execution and deployment of smart contracts provides features to fetch the bytecode so that it can be integrated and ported to other toolkits and programming frameworks.

Vyper programming language for smart contracts

Vyper is another very high performance programming language that is used for the development of smart contracts. Smart contracts using Vyper can be programmed on standalone systems or on the cloud.

The key features of the Vyper programming language that give it an edge over other languages are:

- Less complex and easy to program
- Successor of Serpent language
- Fast execution speed
- Integration of bounds and overflow checking for security and integrity
- Object-oriented programming based language with high security

Smart contracts can be written and compiled on the online compilation platform of Vyper (<https://etherscan.io/vyper>). They can be executed on this URL, and the bytecode can be taken by the programmers.

Vyper can be integrated with Remix IDE as the latter has a Vyper plugin.

Since blockchain based software development is a relatively new field, there are many questions to be answered, such as how to optimise resources as blockchains use a lot of energy and power. Data security and integrity problems can develop when using blockchain and other decentralized apps since data is duplicated on massive devices. The efficiency of blockchain-based applications can be improved by the creation and use of advanced and optimised algorithms. **END** 🐧

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